

SHANGHAI HONGQIAO INTL, China

WMO: 583670

Lat: 31.198N

Lon: 121.336E

Elev: 3

StdP: 101.29

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.5	-1.1	-12.2	1.3	3.0	-9.9	1.6	3.3	10.0	3.3	9.1	3.8	2.5	310	0.462

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.9	36.1	27.2	34.9	27.0	33.8	26.8	28.7	33.0	28.0	32.2	27.4	31.5	3.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.8	23.9	30.8	27.0	22.7	30.3	26.2	21.6	29.9	92.3	33.1	89.4	32.9	87.0	31.8	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.1	8.1	7.2	DB	-4.5	38.2	1.5	1.3	-5.6	39.1	-6.5	39.9	-7.4	40.6	-8.5	41.5	(4)
			WB	-6.0	29.5	1.6	0.8	-7.2	30.0	-8.1	30.5	-9.0	30.9	-10.2	31.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.7	5.1	6.8	10.9	16.5	21.6	25.0	29.6	29.3	25.2	20.0	14.0	7.6
	DBStd	9.01	3.42	3.73	3.80	3.54	2.74	2.67	2.74	2.38	2.77	2.90	3.84	3.70
	HDD10.0	397	157	103	33	2	0	0	0	0	0	0	9	92
	HDD18.3	1533	411	324	231	76	4	0	0	0	0	17	138	333
	CDD10.0	3206	5	13	62	196	361	452	609	599	456	308	128	17
	CDD18.3	1302	0	0	2	21	107	202	351	341	206	67	7	0
	CDH23.3	12913	0	1	13	132	627	1520	4481	4188	1687	247	16	0
	CDH26.7	5507	0	0	2	30	168	514	2294	1968	498	33	1	0
Wind	WSAvg	3.7	3.6	3.7	4.0	3.9	3.9	3.5	3.9	4.1	3.7	3.4	3.4	3.4
Precipitation	PrecAvg	1291	70	71	94	87	99	218	146	208	115	70	63	50
	PrecMax	1717	207	161	245	161	216	720	300	464	261	298	133	122
	PrecMin	871	15	19	29	37	21	27	27	19	15	6	9	0
	PrecStd	225	38	41	52	33	47	135	78	96	61	67	41	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.8	21.8	25.0	30.0	32.9	35.2	38.9	38.0	34.2	29.8	25.2	18.9
		MCWB	13.2	15.3	16.3	20.2	21.5	27.0	27.1	27.6	26.5	22.8	18.9	14.4
	2%	DB	14.2	17.0	21.9	26.9	30.1	33.1	36.9	35.9	32.2	27.1	23.0	16.8
		MCWB	10.7	12.4	14.5	19.2	20.7	25.9	27.3	27.3	25.7	20.9	17.6	12.9
	5%	DB	12.1	14.2	19.2	24.2	28.2	31.2	35.2	34.2	30.9	25.9	21.2	14.9
		MCWB	9.1	10.4	13.3	17.6	20.0	24.7	27.1	27.0	24.9	19.9	17.0	11.4
	10%	DB	10.2	12.2	17.0	22.2	26.9	29.8	34.1	33.1	29.2	24.8	19.9	13.2
		MCWB	7.7	8.7	12.7	16.3	19.4	23.8	27.1	26.7	23.8	19.4	15.9	10.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.2	16.2	18.1	22.6	25.0	28.0	29.6	29.7	28.9	25.1	21.4	16.6
		MCDB	16.6	20.2	21.5	26.8	30.0	32.8	34.1	33.8	31.6	27.9	23.4	18.2
	2%	WB	11.8	13.3	16.1	20.8	23.5	27.0	28.7	28.6	27.0	23.0	19.6	13.8
		MCDB	13.6	15.8	19.6	24.4	27.0	31.5	33.4	32.8	30.0	25.4	21.7	15.7
	5%	WB	9.9	11.3	14.8	19.3	22.3	26.2	28.1	27.9	26.2	21.7	18.0	12.2
		MCDB	11.2	14.0	18.1	22.7	25.6	29.8	32.7	31.7	29.2	24.1	20.1	14.2
	10%	WB	8.0	9.6	13.1	17.8	21.3	25.3	27.5	27.3	25.3	20.6	16.6	10.5
		MCDB	9.7	11.5	16.3	21.3	24.2	28.3	32.0	31.1	28.5	23.2	19.0	12.6
Mean Daily Temperature Range	5% DB	MDBR	6.6	6.9	7.7	8.3	8.0	6.5	6.9	6.3	6.5	7.2	7.4	7.2
		MCDBR	9.3	10.5	11.5	11.5	10.7	9.2	8.6	7.7	7.4	8.2	8.7	9.1
		MCWBR	6.0	6.3	6.2	5.9	4.5	3.6	3.0	2.8	2.9	3.6	4.4	5.7
	5% WB	MCDBR	7.4	9.3	10.0	9.7	8.1	7.8	7.8	7.2	6.5	6.4	6.3	6.7
		MCWBR	5.8	6.5	6.6	6.1	4.7	3.7	3.3	3.1	3.2	3.9	4.4	5.5
Clear-Sky Solar Irradiance	taub	0.598	0.638	0.671	0.679	0.663	0.717	0.685	0.660	0.614	0.612	0.620	0.590	
	taud	1.664	1.611	1.574	1.616	1.681	1.619	1.715	1.750	1.816	1.739	1.677	1.692	
	Ebn at Noon	594	622	644	666	681	643	661	671	681	635	571	572	
	Edh at Noon	206	238	263	261	246	261	236	226	204	206	202	192	
All-Sky Solar Radiation	RadAvg	2.25	2.64	3.73	4.40	4.74	4.15	5.28	4.87	3.96	3.38	2.49	2.22	
	RadStd	0.29	0.51	0.31	0.50	0.52	0.61	0.63	0.65	0.39	0.45	0.30	0.32	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.38	+0.46	N/A	+0.66	N/A	N/A	N/A	-109	+134	+75
(47)	Regional (10 neighbors)	+0.29	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+86	N/A

Nomenclature: See separate page